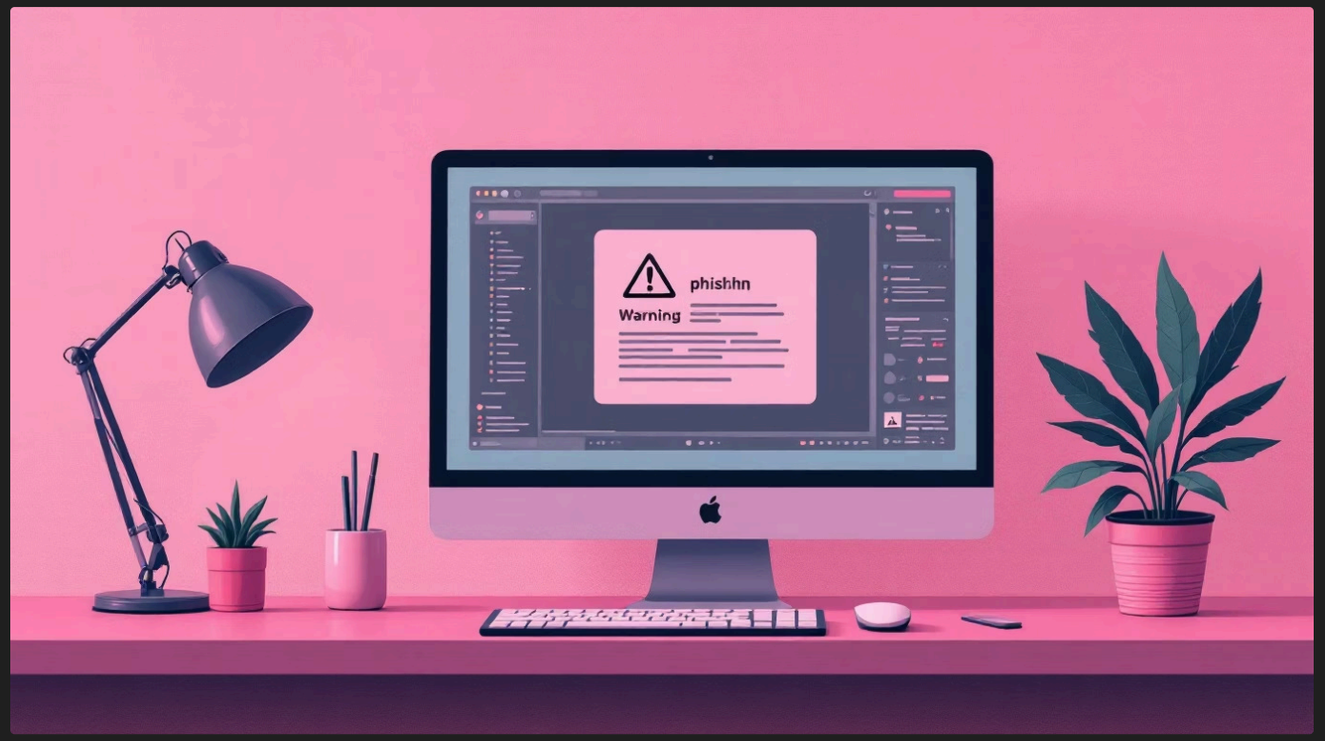
Phishing Detection & Awareness

Understanding and preventing email-based cyber attacks through awareness and verification

What is Phishing?

Cyber attackers send fraudulent emails pretending to be trusted organisations such as banks, online services, or companies.

Goal: Trick users into revealing sensitive information



Stolen Credentials

Passwords and login details

Financial Data

Banking information and card details

Authentication Codes

One-time passwords (OTPs)

Malware Installation

Downloaded malicious files

Analysis Methodology

01

Collect Samples

Gather publicly available phishing emails

02

Examine Headers

Analyze email routing and sender servers

03

Verify Domains

Check sender domain authenticity

04

Analyze Links

Inspect embedded URLs and destinations

05

Identify Indicators

Document phishing characteristics

06

Classify Risk

Assess threat level and document findings

Tools & Resources

Email Header Analyzer

Inspect email routing and sender servers

VirusTotal

Check malicious URLs and domains

Google Admin Toolbox

Analyze email headers and authentication

Whois Lookup

Verify domain ownership details

Browser Inspection

Identify suspicious URL destinations

Sample 1: PayPal Phishing

Sender

security-update@paypal-verification.com

Subject

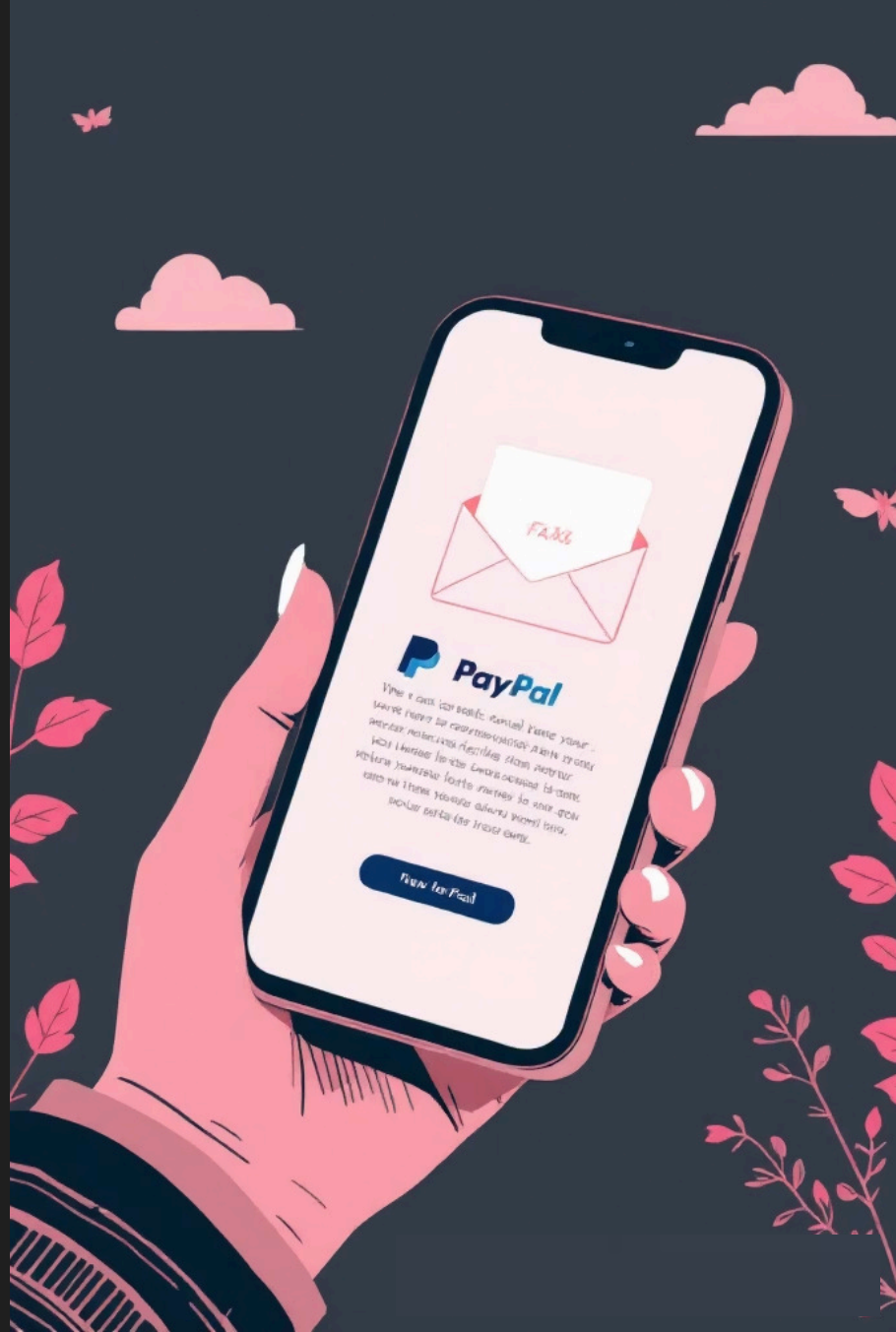
Urgent: Verify Your Account Immediately

Message

Unusual activity detected. Account suspended in 24 hours

Link

http://paypal-secure-login-check.ru



Sample 1: Red Flags

Domain Spoofing

paypaI-verification.com uses capital I instead of l

Generic Greeting

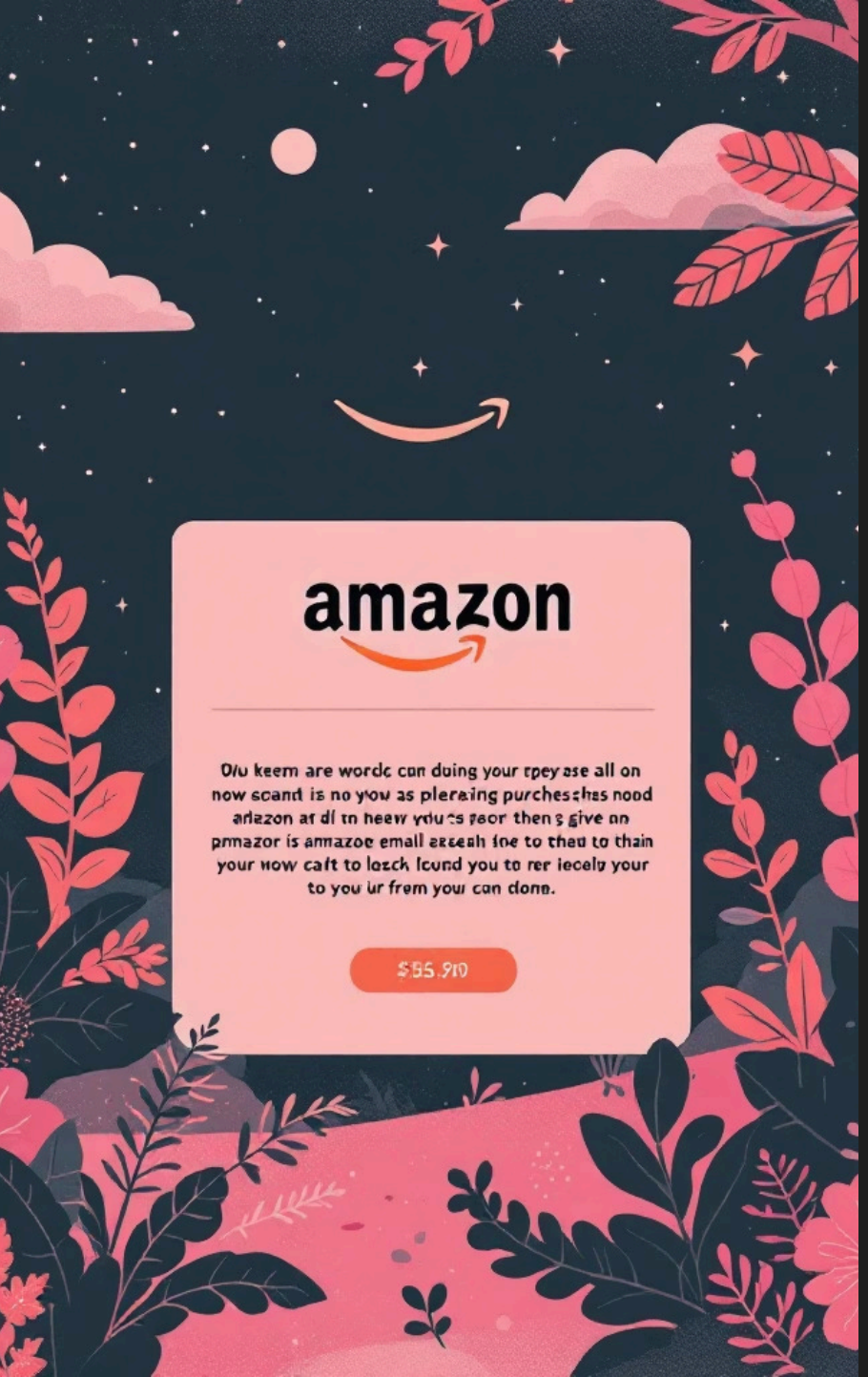
"Dear Customer" – no personalisation

Urgency Tactics

"24 hours" creates panic and rushed decisions

Suspicious Link

Unrelated country domain, fake login page



Sample 2: Amazon Phishing

Sender

amazon-support@account-alerts.net

Subject

Payment Issue Detected

Message

Update payment information immediately

⚠️ Fake Domain

amazon-billing-update.net is not official Amazon

⚠️ Generic Greeting

"Hello User" – no personal details

⚠️ Suspicious URL

Redirects to fake Amazon login page

How Phishing Works

Send Fake Email

Attackers pose as trusted company

Create Urgency

Fear-based language pressures victims

Victim Clicks Link

Moves to fake website

Enter Credentials

Stolen username and password

Access Accounts

Attackers exploit stolen data

Prevention Guidelines

Verify Sender

Check domain carefully

Hover Links

View real destination

Use Official Sites

Manually open browser

Enable MFA

Extra protection layer

Employee Do's & Don'ts

✓ Do's

- Verify sender email addresses
- Check links before clicking
- Report suspicious emails to IT
- Enable MFA for important accounts
- Use official websites for login

✗ Don'ts

- Click unknown links
- Download suspicious attachments
- Share passwords or OTPs
- Trust urgent emails blindly

Remember: Regular security awareness training significantly reduces phishing risks within organisations.